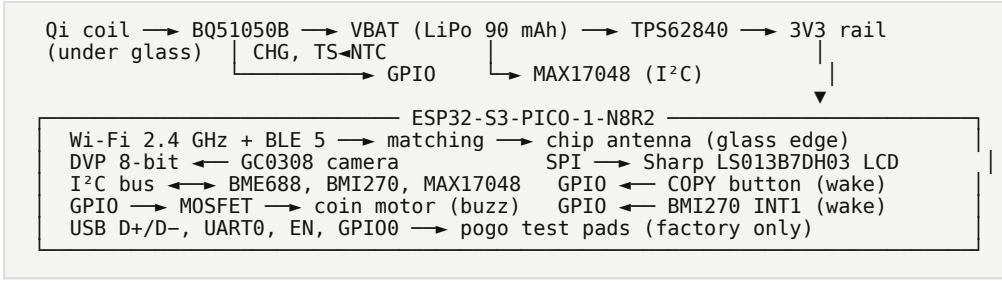
```
================================================================================
LOT-01 — ELECTRICAL DESIGN
================================================================================
DOCUMENT    HW-04 / ELEC
ISSUE       2026.11.12 · REV A (preliminary — schematic capture is session 02+)
PCB         36 × 36 mm, 6-layer HDI 0.6 mm (EVT) · 4-layer 0.4 mm (DVT)
================================================================================
```

# 1 · BLOCK DIAGRAM




# 2 · PIN MAP (preliminary)

Avoid strapping pins GPIO0 / 3 / 45 / 46 for anything that is driven at boot. N8R2 has quad PSRAM, so GPIO 33–37 stay free.

| Function | Signal | GPIO | Notes |
|---|---|---|---|
| Camera | XCLK | 15 | 20 MHz via LEDC |
| | PCLK | 13 | |
| | VSYNC / HREF | 6 / 7 | |
| | D0–D7 | 11, 9, 8, 10, 12, 18, 17, 16 | |
| | SIOD / SIOC | 4 / 5 | SCCB, separate from sensor I²C |
| | PWDN | 39 | camera off by default (non-RTC is fine) |
| LCD (SPI) | SCLK / MOSI / CS | 36 / 35 / 34 | CS is active-high on Sharp |
| | EXTCOMIN | 33 | 1 Hz toggle from RTC (or software) |
| | DISP | 37 | display on/off |
| Sensor I²C | SDA / SCL | 1 / 2 | BME688 0x76, BMI270 0x68, MAX17048 0x36 |
| IMU wake | INT1 | 21 | RTC-capable, wake on tap/lift |
| COPY button | BTN | 14 | RTC GPIO → wakes from deep sleep (ext1), 10 k pull-up |
| Motor | MOT | 40 | via N-MOSFET |
| Charger status | CHG | 41 | BQ51050B open-drain |
| Fuel gauge alert | ALRT | 42 | |
| Factory | U0TXD / U0RXD | 43 / 44 | pogo pads |
| | USB D− / D+ | 19 / 20 | pogo pads, flashing + logs |

**Open item (pin map).** Deep-sleep wake needs RTC GPIOs (0–21 on ESP32-S3). The camera bus takes most of them, so camera PWDN sits on GPIO 39 and COPY takes the freed RTC pin GPIO 14; BMI270 INT1 uses RTC GPIO 21. Some ESP32-S3-PICO-1 pins are not bonded out or are used internally: check every row against the PICO-1 datasheet pin list during schematic capture. Treat this table as a starting point.

## 3 · POWER BUDGET

| State | Current | Time / day | mAh / day |
|---|---|---|---|
| Deep sleep + LCD static + IMU low-power | ~ 15 µA | ~ 23.9 h | 0.36 |
| Wake + Wi-Fi + HTTPS poll | ~ 100 mA avg × 3 s | 288 polls (5 min) | 24.0 |
| BME688 reading (BSEC2 LP, 3 s) | low | continuous LP | 1.0 |
| COPY + frame upload (10/day) | ~ 180 mA × 5 s | 10 × | 2.5 |
| Buzz (10 × 300 ms) | 60 mA | | 0.05 |
| **Total** | | | **≈ 28** |

90 mAh cell → **≈ 3 days** at 5-minute polls; **≈ 6 days** at 10-minute polls. The server returns `next_poll_s`, so it can slow polling at night and speed it up when a message is expected (HW-06).

## 4 · CHARGING (Qi)

- BQ51050B: Qi v1.2 receiver and Li-ion charger in one chip, 4.20 V float.
- Set charge current ≈ 45 mA (0.5 C) with the ILIM resistor. Small cell, small coil, low heat.
- 10 k NTC on the battery → TS pin. Stop charge above 45 °C.
- CHG pin → GPIO so the screen can show "Charging." and "Charged."
- Coil location: under the LCD, ferrite toward the PCB. **EVT test:** measure charge efficiency and LCD temperature with the coil under the LCD. Fallback: smaller coil (Ø 20 mm) shifted below the LCD outline.

## 5 · SENSORS ("AI-grade, off-the-shelf")

| Sensor | Measures | AI part |
|---|---|---|
| BME688 | Temperature, humidity, pressure, VOC/gas | Bosch BSEC2 + BME AI-Studio: train gas classes (e.g. *coffee*, *fresh air*, *cooking*) and run them on-device |
| BMI270 | Motion, tap, lift, orientation | Built-in gesture/activity features; wakes the MCU on lift |
| GC0308 | VGA image | Frames go to LOT for AI description, or ESP-DL on-device |
| ESP32-S3 | — | Vector instructions for small on-device models (ESP-DL) |

Placement: BME688 on the bottom-edge of the PCB, thermally isolated by a routed slot, behind the vent. Keep it ≥ 8 mm from the ESP32-S3 and the charger IC.

## 6 · RF

- Chip antenna at the top edge, on the glass side of the PCB, with a ≥ 5 × 8 mm copper keep-out on all layers.
- Pi matching network (3 × 0201) — tune on EVT bodies with a VNA, inside the closed steel case.
- Pass criteria: RSSI ≥ −70 dBm at 5 m line of sight from a home AP.

## 7 · TEST PADS (factory)

Six pogo pads on the PCB bottom, reachable before the battery is fitted: VBAT, GND, EN, GPIO0, USB D−, USB D+. Flashing, serial number, and self-test happen here (HW-07).